

[illegible]

# SPRING AI

## GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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### Note

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## SPEECH-TO-TEXT

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### Note

[illegible]

```

...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-starter-model-openai</artifactId>
</dependency>
...

```

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
```

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```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

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## Controller di trascrizione

```

@RestController
@RequestMapping("/api")
public class AudioController {

    ...

    @GetMapping("/transcribe-options")
    String transcribeWithOptions(@Value("classpath:filename.mp3") Resource audioFile) {
        var audioTranscriptionResponse = openAiAudioTranscriptionModel.call(new AudioTranscriptionPrompt(
            audioFile, OpenAiAudioTranscriptionOptions.builder()
                .prompt("Topics")
                .language("it")
                .temperature(0.5f)
                .responseFormat(OpenAiAudioApi.TranscriptResponseFormat.VTT).build()));
        return audioTranscriptionResponse.getResult().getOutput();
    }
}

```

- ➔ `AudioTranscriptionPrompt` incapsula la risorsa audio e le opzioni di trascrizione (`OpenAiAudioTranscriptionOptions`)
  - ➔ `.prompt` fornisce al LLM contesto riguardo gli argomenti trattati nel *file* audio
  - ➔ `.language` evita al LLM l'attività di *language guessing*
  - ➔ `.responseFormat` per calibrare il formato di *output* al contesto applicativo di utilizzo
- ⚠ con `call(audioFile)`, Spring AI si occupa di creare un *bean* `OpenAiAudioTranscriptionOptions` con i valori di *default*

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```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
      audio:
        transcription:
          model: whisper-1           # whisper-1, gpt-4o-transcribe, gpt-4o-mini-transcribe
          response-format: json     # json, text, srt, verbose_json, vtt
          prompt: Topics            # Should match the audio language
          language: it             # ISO-639-1 format to improve accuracy and latency
          temperature: 0           # [0, 1]. 0 means automatically adjusting temperature
          ...
```

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```

AUDIO_PATH="path/to/sample.mp3"
MIME_TYPE=$(file -b --mime-type "${AUDIO_PATH}")
NUM_BYTES=$(wc -c < "${AUDIO_PATH}")
DISPLAY_NAME=AUDIO

tmp_header_file=upload-header.tmp

# Initial resumable request defining metadata.
# The upload url is in the response headers dump them to a file.
curl "https://generativelanguage.googleapis.com/upload/v1beta/files" \
  -H "x-goog-api-key: $GEMINI_API_KEY" \
  -D upload-header.tmp \
  -H "X-Goog-Upload-Protocol: resumable" \
  -H "X-Goog-Upload-Command: start" \
  -H "X-Goog-Upload-Header-Content-Length: ${NUM_BYTES}" \
  -H "X-Goog-Upload-Header-Content-Type: ${MIME_TYPE}" \
  -H "Content-Type: application/json" \
  -d '{"file": {"display_name": "${DISPLAY_NAME}"}}' 2> /dev/null

upload_url=$(grep -i "x-goog-upload-url: " "${tmp_header_file}" | cut -d" " -f2 | tr -d "\r")
rm "${tmp_header_file}"

# Upload the actual bytes.
curl "${upload_url}" \
  -H "Content-Length: ${NUM_BYTES}" \
  -H "X-Goog-Upload-Offset: 0" \
  -H "X-Goog-Upload-Command: upload, finalize" \
  --data-binary "@${AUDIO_PATH}" 2> /dev/null > file_info.json

file_uri=$(jq ".file.uri" file_info.json)
echo file_uri=$file_uri

```

### Note

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- 1 Utilizzo Spring Web puro (RestController, ResponseEntity, HttpEntity, ...)
- 2 Utilizzo Spring AI (STT attraverso ChatClient)

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➔ Servizio *speech-to-text* Gemini

- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 Test delle funzionalità con Postman/Insomnia

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```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

}
```

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```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;

    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content());

    }

    ...
}
```

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@RequestMapping("/gemini/multi-modality")
public class GeminiMultimodalityController {

    private final MultiModalityService multiModalityService;

    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }

    /**
     * Endpoint per la trascrizione di file audio in testo (Speech-to-Text).
     * Utilizza un file audio predefinito dal classpath e ne estrae la trascrizione testuale
     * tramite i servizi di riconoscimento vocale di Gemini.
     *
     * @param audioFile risorsa audio dal classpath (file WAV predefinito)
     * @return Answer oggetto contenente la trascrizione del file audio
     */
    @PostMapping("/stt")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {

        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }
}
```

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